CST283 Data Structures and Algorithms
Project 1 – Guidelines

Project 1 – CST283.1 Data Structure Playground (Structured Selection Model) targets LO1, LO2, LO3 (foundations).

Purpose

This project presents a curated set of implementations and comparisons designed to demonstrate an understanding of:

- the purpose and use of data structures,
- the behavior of their core operations, and
- how performance scales with increasing input size, combining conceptual Big-O analysis with basic empirical timing results.

Instructions on Deliverables

- You may extract codes from your weekly class activities to populate your submission for this project
- Submit Google colab file – Codes and documentation in the same file.
- Submitting separate documentation as PDF/word is not required as long as your colab file includes sufficient documentation as 'text' part of the colab file.
- Live demonstration of your code and Q&A based on your code/documentation in week 9

Step 1 — Select Structures from the following Categories

You must choose **2 categories** from the table below, and within each chosen category, choose **2 structures**. Your submission must therefore cover: **2 categories × 2 structures = 4 structures in total.**

Sl.No.	Category	Structures
1	**Linear / Fundamental Structures *(Ch. 3–5)***	• Array / Dynamic Array (Ch.3) • Stack (Ch.4) • Queue / Circular Queue / Deque (Ch.4) • Linked List / Doubly Linked List / Positional List (Ch.5)
2	**Hierarchical Structures *(Trees) (Ch.6)***	• General Tree (Ch.6) • Binary Tree (Ch.6) • Expression Tree (Ch.6)

		• Heap / Binary Heap (Ch.6 & used again in Ch.10)
3	**Associative / Map-Based Structures *(Ch.8)***	• Map ADT / Dictionary (Ch.8) • Unsorted Table Map (Ch.8) • Hash Table (Ch.8) • Sorted Map (Ch.8) • Skip List (Ch.8) • Set / Multiset / Multimap (Ch.8)

Step 2 — Required Demonstrations (For Each Selected Structure)

For **each of your 6 selected structures**, include the following four parts.

Sl. No.	Task	Description
1	**Implementation**	• A clean Python **class or function** (appropriate to the structure) • **Original comments** explaining key logic (in your own words) • Clear variable names and readable structure
2	**Operations**	Demonstrate at least: • **Insert** • **Delete** *(where applicable)* • **Search** • **Traversal** *(for trees; for graphs use BFS/DFS as traversal)* **Note:** Some structures don't naturally support all operations (e.g., arrays don't "delete" without shifting; tries don't "traverse" the same way as trees). If an operation is not applicable, explain why briefly and demonstrate the closest equivalent.
3	**Visual Representation**	Include either: • A **diagram** (hand-drawn, digital, or generated), **OR** • A **step-by-step print-based visualisation** You must show:

		• The structure **before** an operation • The structure **after** an operation
4	**Reflection (5–8 lines per structure)**	For each structure, write a short reflection answering: • When is this structure useful? • What are its weaknesses? • What is the approximate time complexity of core operations? • When would another structure be better?

Step 3 — Build a "Story" Across Weeks (Comparison Thread)

Your documentation should include a simple narrative thread that connects your chosen structures. Use 'text' sections in the Google Colab file for this narrative/documentation.

Examples of good comparison threads:

- "How does **search** performance change from linear structures → tree-based structures?"
- "How does **insertion/deletion cost** differ between arrays and linked structures?"
- "How do **representation choices** affect traversal and memory (e.g., graphs: adjacency list vs matrix)?"

Keep comparisons **foundational**: you may include simple timing experiments and Big-O reasoning, but do not turn this into a full applied optimisation report.

Suggested Submission Layout

- Title and selected categories/data structures
- For each category:
    - Structure A: Implementation → Operations → Visualization → Reflection
    - Structure B: Implementation → Operations → Visualization → Reflection
- Cross-structure comparison summary (maximum one page)
- References (including any external tools or diagrams used)

Deliverables

- A **Google Colab notebook containing well-documented code** for each data structure, including **implemented operations, visualizations, and written reflections** captured in text cells. (Week 8)

Demonstration Guidelines

- Prepare for a demonstration of your code in week 9 (in-class or a time set out by your facilitator).
- Structure of the demonstration would be:
    - Run code for Category A, Structure 1, Q&A
    - Run code for Category A, Structure 2, Q&A
    - Comparative reflection on Category A, Structure 1 vs Structure 2
    - Run code for Category B, Structure 1, Q&A
    - Run code for Category B, Structure 2, Q&A
    - Comparative reflection on Category B, Structure 1 vs Structure 2
    - Comparative reflection of structures in Category A vs Category B